4. Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.

CODING:

```
import math
import random

random.seed(1)

def sigmoid(x):
  return 1 / (1 + math.exp(-x))

def dsigmoid(y):
  return y * (1 - y)

X = [[0, 0], [0, 1], [1, 0], [1, 1]]
Y = [0, 1, 1, 0]

w1 = [[random.uniform(-1, 1) for _ in range(2)]
    for _ in range(4)]
w2 = [random.uniform(-1, 1) for _ in range(4)]
b1 = [random.uniform(-1, 1) for _ in range(4)]
b2 = random.uniform(-1, 1)
lr = 0.5

for epoch in range(20000):
  for x, target in zip(X, Y):
    hidden = [
```

```
            sigmoid(sum(x[j] * w1[i][j] for j in range(2)) + b1[i])
            for i in range(4)
        ]
        output = sigmoid(sum(hidden[i] * w2[i] for i in range(4)) + b2)

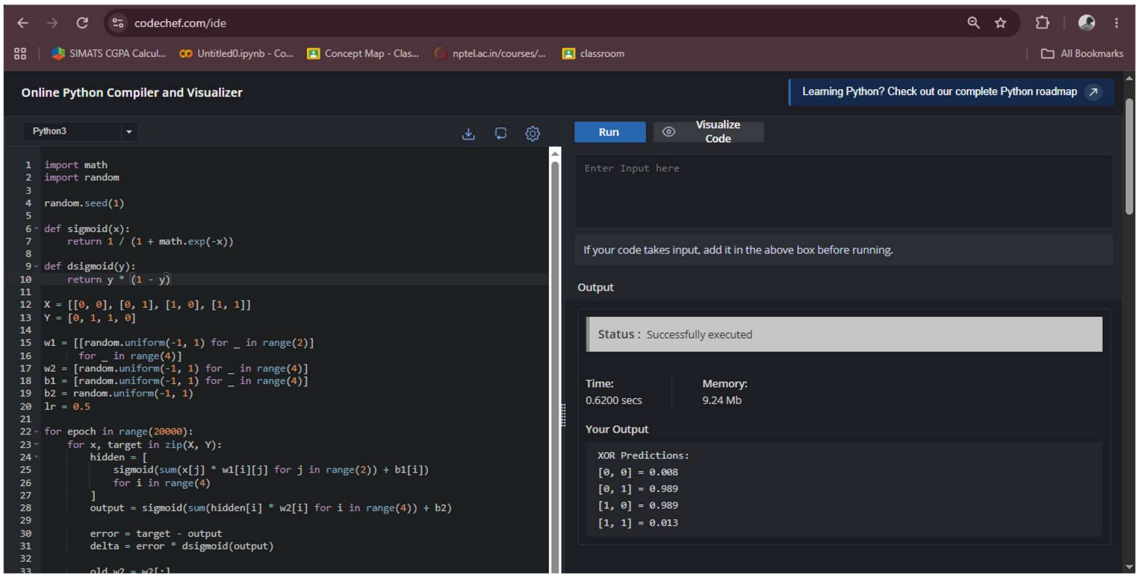
        error = target - output
        delta = error * dsigmoid(output)

        old_w2 = w2[:]
        for i in range(4):
            w2[i] += lr * delta * hidden[i]
        b2 += lr * delta

        for i in range(4):
            d = delta * old_w2[i] * dsigmoid(hidden[i])
            for j in range(2):
                w1[i][j] += lr * d * x[j]
            b1[i] += lr * d

print("XOR Predictions:")
for x in X:
    hidden = [
        sigmoid(sum(x[j] * w1[i][j] for j in range(2)) + b1[i])
        for i in range(4)
    ]
    output = sigmoid(sum(hidden[i] * w2[i] for i in range(4)) + b2)
    print(x, "=", round(output, 3))
```

OUPUT:


codechef.com/ide
SIMATS CGPA Calcul...
Untitled0.ipynb - Co...
Concept Map - Clas...
nptel.ac.in/courses/...
classroom
All Bookmarks
Online Python Compiler and Visualizer
Learning Python? Check out our complete Python roadmap
Python3
Run
Visualize Code
Enter Input here
If your code takes input, add it in the above box before running.
Output
Status : Successfully executed
Time:
0.6200 secs
Memory:
9.24 Mb
Your Output
XOR Predictions:
[0, 0] = 0.008
[0, 1] = 0.989
[1, 0] = 0.989
[1, 1] = 0.013